Chapter 11: Solutions and Colloids

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CHM111

Lecture 11

Part I

The Dissolution Process

Outline of Part I: The Dissolution Process

What Is a Solution?

Outline of Part I: The Dissolution Process

What Is a Solution?

Solutions: A Homogeneous Mixture

Definition: Solution

A **solution** is a homogeneous mixture of two or more substances in which all components are uniformly distributed at the molecular scale.

Solvent & Solute

- **Solvent**: the component present in the greatest amount; determines the physical state of the solution.
- **Solute**: the component(s) dissolved in the solvent.
- An **aqueous solution** has water as the solvent.

Five Properties of Solutions

- Homogeneous throughout
- Physical state matches the solvent
- Dispersed at the molecular scale
- Dissolved solute does not settle
- Composition is variable (within solubility limits)

Solutions: A Homogeneous Mixture

Dissolution of potassium dichromate in water. When solid potassium dichromate ($K_2Cr_2O_7$) is added to water, it dissolves and dissociates into potassium ions (K^+) and dichromate ions ($Cr_2O_7{}^{2-}$). These ions become uniformly dispersed throughout the water, producing a homogeneous orange aqueous solution.

What Makes a Solution Uniform?

- Question
  - Why does some mixing become molecularly uniform instead of remaining separate domains?
- Evidence
  - Salt water appears identical throughout every sampled volume.
  - Microscopy reveals no persistent phase boundaries within a true solution.
- Reasoning
  - Continuous *thermal motion* constantly redistributes discrete particles between neighboring regions.
  - $\therefore$ Uniformity is an entropically favored state because dispersed particles can occupy many more microscopic configurations than clustered particles, even though attractive interactions may energetically favor clustering.
- Takeaway
  - *Molecular dispersion* defines a solution.

Why Does Uniform Mixing Persist?

- Question
  - Why does molecular uniformity persist instead of separating after mixing stops?
- Evidence
  - Many solutions remain unchanged for months without stirring.
  - Individual molecules continue moving despite the stable appearance.
- Reasoning
  - Mixing lowers the system's *free energy* when solvent–solute interactions compete successfully with like-particle attractions.
  - $\therefore$ The homogeneous state becomes an equilibrium, not merely a transient arrangement.
- Takeaway
  - Stable solutions are equilibrium states.

Why Is One Component Called the Solvent?

- Question
  - If mixing is symmetric, why distinguish solvent from solute?
- Evidence
  - Exchanging labels leaves the same molecular arrangement unchanged.
  - The more abundant component is conventionally named the *solvent*.
- Reasoning
  - Intermolecular forces act between particle pairs, independent of human naming.
  - $\therefore$ Solvent and solute labels organize composition rather than describe different physics.
- Takeaway
  - Names record amount, not mechanism.

When Does the Asymmetry Become Physical?

- Question
  - When does the solvent label acquire predictive physical significance?
- Evidence
  - Vapor pressure changes mainly track the majority liquid component.
  - Many solution properties depend on the solvent's escaping molecules.
- Reasoning
  - Surface molecules entering the vapor are overwhelmingly supplied by the majority component.
  - $\therefore$ The concentration asymmetry enters laws such as Raoult's through particle availability.
- Takeaway
  - Concentration creates observable asymmetry.

Why Does a Solution Not Settle Out?

- Question
  - Why is "does not settle out" a thermodynamic statement rather than merely slow motion?
- Evidence
  - True solutions remain uniform indefinitely under constant conditions.
  - Suspensions eventually separate into distinct regions.
- Reasoning
  - Equilibrium favors dispersed particles, whereas suspended aggregates occupy a higher-energy, nonequilibrium state.
  - $\therefore$ Settling reflects movement toward equilibrium, not simply insufficient mixing speed.
- Takeaway
  - Equilibrium determines long-term stability.

Types of Solutions

Solutions exist in all phases — not just liquid:

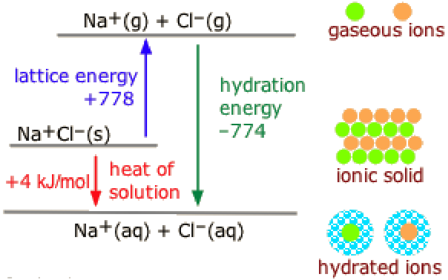
Solute	**Solvent**	**Example**
Gas (O_2)	Gas (N_2)	Air
Gas (CO_2)	Liquid (H_2O)	Soft drinks
Gas (H_2)	Solid (Pd)	Hydrogen in palladium
Liquid (H_2O)	Liquid (C_3H_8O)	Rubbing alcohol
Solid (NaCl)	Liquid (H_2O)	Saltwater
Solid (Zn)	Solid (Cu)	Brass

The solvent is the component present in the greatest concentration; everything else is a solute.

The Energetics of Dissolution

Dissolution consists of two ideas:

- **Molecular Process:**
  1. separate solute
  2. separate solvent
  3. mix to form hydrated ions
- **Energy:**
  - energy absorbed to separate particles (Endothermic)
  - energy released during hydration (Exothermic)



$$\Delta H_{\text{soln}} = \Delta H_{\text{latt}} + \Delta H_{\text{hyd(cation)}} + \Delta H_{\text{hyd(anion)}}$$

Even if $\Delta H_{\text{soln}} > 0$ (endothermic), dissolution can still be **spontaneous** if entropy increases enough. Example: NH_4NO_3 dissolving (instant cold packs).

Driving Forces for Dissolution

Two thermodynamic driving forces promote dissolution:

1. Decrease in Energy

If solute–solvent interactions release more energy than separation requires, dissolution is energetically favorable ($\Delta H_{\text{soln}} < 0$).

2. Increase in Entropy

Mixing disperses matter and increases system disorder. This thermodynamic tendency favors dissolution even when it is endothermic.

Examples

- **NaOH** in water: highly exothermic — solvation forces exceed lattice energy.
- $\mathbf{NH_4NO_3}$ in water: endothermic, yet spontaneous — driven by entropy (instant cold packs).
- $\mathbf{CaCO_3}$: lattice energy far exceeds solvation energy — essentially insoluble.

Why Can Dissolution Be Split Into Imaginary Steps?

- Question
  - Why can dissolution be analyzed using steps that never occur separately?
- Evidence
  - Measured enthalpy change is identical regardless of experimental pathway.
  - Different reaction routes reach the same final state.
- Reasoning
  - $\because$ Energy depends only on initial and final molecular arrangements, not the path taken.
  - $\therefore$ Imaginary separation and mixing steps legitimately reveal individual energy contributions.
- Takeaway
  - State functions permit mechanistic energy decomposition.

What Determines Whether Dissolution Occurs?

- Question
  - Why do some dissolutions occur despite absorbing heat?
- Evidence
  - Some salts dissolve while cooling the surrounding solution.
  - Spontaneous dissolutions show both positive and negative enthalpy changes.
- Reasoning
  - $\because$ Spontaneity balances lower potential energy with greater molecular accessibility, quantified by $\Delta G = \Delta H - T\Delta S$.
  - $\therefore$ Dissolution occurs when the combined effect makes ΔG negative.
- Takeaway
  - *Free energy* balances energetic and configurational effects.

Why Does Particle Dispersion Increase Entropy?

- Question
  - Why does mixing particles increase *entropy*?
- Evidence
  - Mixed particles occupy far more (distinguishable) arrangements than separated particles.
  - Entropy measurements increase during many dissolutions.
- Reasoning
  - $\because$ More accessible molecular arrangements make any single ordered arrangement overwhelmingly less probable.
  - $\therefore$ Dispersion naturally favors higher entropy, often rivaling typical interaction energies.
- Takeaway
  - More accessible arrangements increase spontaneous dispersion.

When Can Entropy Overcome Endothermic Dissolution?

- Question
  - When can entropy outweigh the energy required to separate particles?
- Evidence
  - Some salts dissolve only above characteristic temperatures.
  - The entropy contribution grows with absolute temperature.
- Reasoning
  - $\because$ Higher temperature increases the energetic significance of accessing many molecular arrangements.
  - $\therefore$ A positive ΔH can be overcome when $T\Delta S$ becomes larger.
- Takeaway
  - Temperature amplifies entropy's thermodynamic influence.

Why Do Lattice Breaking and Hydration Have Opposite Signs?

- Question
  - Why does separating a crystal require energy while hydration releases it?
- Evidence
  - Ionic crystals resist separation.
  - Water molecules orient around dissolved ions.
- Reasoning
  - $\because$ Breaking close electrostatic attractions raises potential energy, whereas oriented ion–dipole attractions lower it.
  - $\therefore$ *Lattice energy* is endothermic and *hydration* is usually exothermic.
- Takeaway
  - Interaction geometry determines energy change.

Ideal Solutions

Definition: Ideal Solution

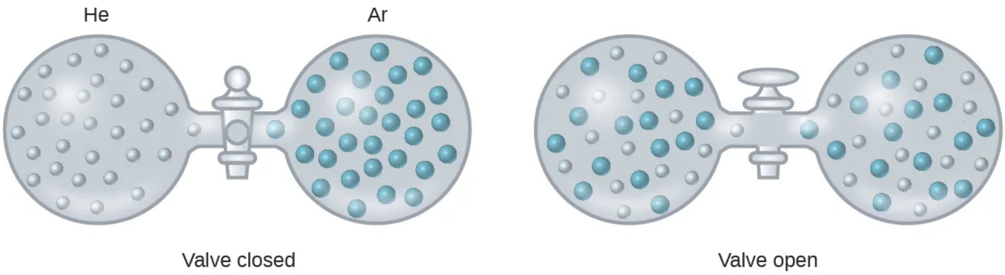
A solution in which the solute–solvent interactions are essentially equal to the solute–solute and solvent–solvent interactions, so that $\Delta H_{\text{soln}} \approx 0$.

Examples of Ideal (or Near-Ideal) Solutions

- Helium and argon gas mixtures
- Methanol and ethanol
- Pentane and hexane

These mixtures form spontaneously due to the **increase in entropy** — even without an energy change. The intermolecular forces between the mixed species are virtually identical to those in the pure components, so mixing is essentially free energetically.

Spontaneous Mixing of He and Ar



Formation of an ideal gaseous solution. Opening the stopcock allows helium and argon atoms to diffuse into both containers, producing a homogeneous mixture. Because the gases have negligible intermolecular attractions, mixing occurs with essentially no energy change and is driven by the increased dispersal of the gas particles.

Why Does Removing a Partition Increase Entropy?

- Question
  - Why does removing a partition spontaneously mix two different gases?
- Evidence
  - Helium and argon spread uniformly after the partition is removed.
  - No measurable temperature change accompanies ideal-gas mixing.
- Reasoning
  - $\because$ Each molecule can explore a larger volume, creating many additional distinguishable molecular arrangements.
  - $\therefore$ The system overwhelmingly occupies mixed states, increasing *entropy* despite unchanged energy.
- Takeaway
  - More accessible arrangements drive spontaneous mixing.

Why Is Entropy of Mixing Zero for Identical Gases?

- Question
  - Why does mixing identical gases produce no entropy increase?
- Evidence
  - Removing a partition between identical gases produces no observable macroscopic change.
  - Exchanging identical molecules creates no experimentally distinguishable arrangement.
- Reasoning
  - $\because$ Entropy counts physically distinguishable molecular arrangements, not merely exchanged particle labels.
  - $\therefore$ Only distinguishable species gain additional accessible microstates upon mixing.
- Takeaway
  - Distinguishability determines mixing entropy.

Why Is Mixing an Ideal Solution Energetically Neutral?

- Question
  - What molecular condition makes ΔH_{mix} essentially zero?
- Evidence
  - Ideal mixtures show negligible heat absorption or release.
  - Their properties closely follow composition-based predictions.
- Reasoning
  - $\because$ Replacing A–A and B–B neighbors with A–B neighbors leaves average intermolecular potential energy unchanged.
  - $\therefore$ Mixing changes accessible arrangements without changing interaction energy.
- Takeaway
  - Equal interactions isolate entropy as the driving force.

Why Do Oil and Water Resist Mixing?

- Question
  - If mixing increases entropy, why do oil and water separate?
- Evidence
  - Oil droplets coalesce in water instead of remaining uniformly dispersed.
  - Water molecules become highly ordered around isolated nonpolar molecules.
- Reasoning
  - $\because$ Maintaining water's hydrogen-bond network restricts nearby molecular orientations, reducing water's entropy.
  - $\therefore$ This entropy loss can exceed the entropy gained by mixing, preventing spontaneous mixing.
- Takeaway
  - Total entropy change determines mixing.

Check Your Understanding

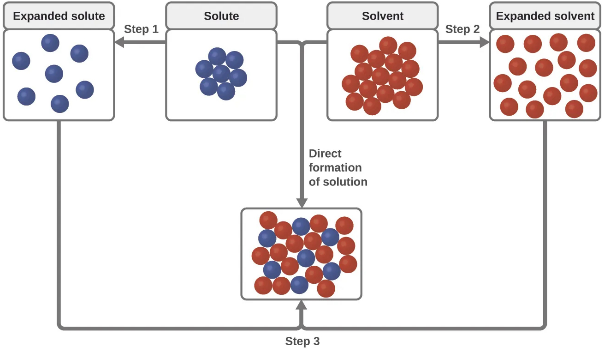
When ammonium nitrate (NH_4NO_3) dissolves in water, the solution becomes very cold.

(a) Is this dissolution **exothermic** or **endothermic**?

(b) Does this mean it will *not* dissolve spontaneously? Explain briefly.

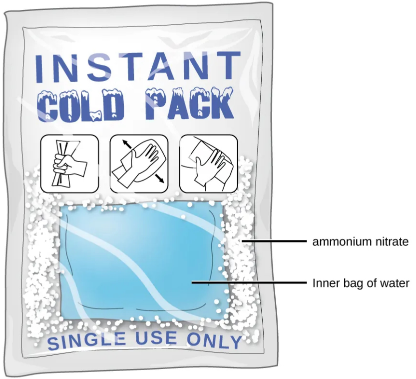
Come to lecture to see the answer.

Energy Changes During Solution Formation



Energy changes during solution formation. Dissolution can be viewed as a three-step process: (1) separating solute particles, (2) separating solvent particles, and (3) forming new solute-solvent interactions (solvation). The first two steps require energy to overcome intermolecular attractions, whereas solvation releases energy. The overall energy change depends on the balance between these opposing processes.

Endothermic Dissolution and Spontaneous Solution Formation



An instant cold pack demonstrates endothermic dissolution. Breaking the inner water pouch allows ammonium nitrate (NH_4NO_3) to dissolve. Because the dissolution process absorbs more energy than it releases, heat is drawn from the surroundings, causing the pack to become cold while the salt dissolves spontaneously.

Part II

Electrolytes

Outline of Part II: Electrolytes

Electrolytes and Nonelectrolytes

Outline of Part II: Electrolytes

Electrolytes and Nonelectrolytes

Electrolytes vs. Nonelectrolytes

Electrolyte

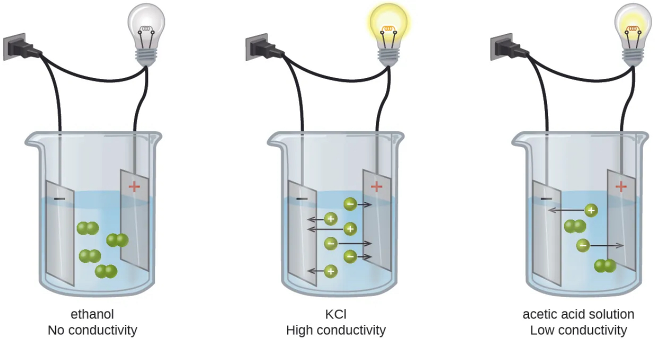
A substance that produces **ions** when dissolved in water, enabling the solution to conduct electricity.

Nonelectrolyte

A compound that dissolves without producing ions — the solution does **not** conduct electricity.

Examples: ethanol (C_2H_5OH), sucrose, glucose

Electrical conductivity is directly proportional to dissolved ion concentration.



Solutions of nonelectrolytes such as ethanol do not contain dissolved ions and cannot conduct electricity. Solutions of electrolytes contain ions that permit the passage of electricity. The conductivity of an electrolyte solution is related to the strength of the electrolyte.

Strong and Weak Electrolytes

Strong Electrolyte

Ion-producing process is essentially **100% complete**.

Examples: NaCl, KCl, HCl, HNO_3, NaOH

Weak Electrolyte

Only a **small fraction** of dissolved molecules produce ions (partial ionization).

Examples: acetic acid (CH_3COOH), NH_3

$$\mathrm{NaCl}(s) \xrightarrow{H_2O} \mathrm{Na}^+(aq) + \mathrm{Cl}^-(aq)$$

$$\mathrm{CH_3COOH}(aq) \rightleftharpoons \mathrm{H}^+(aq) + \mathrm{CH_3COO}^-(aq)$$

The double arrow ($\rightleftharpoons$) signals **equilibrium**: weak electrolytes only partially ionize. Strong electrolytes use a single forward arrow ($\rightarrow$) because ionization is essentially complete.

Why Does Conductivity Reveal Ion Formation?

- Question
  - Why does an ionic solution conduct while a molecular solution does not?
- Evidence
  - NaCl(aq) conducts strongly; sugar(aq) shows negligible conductivity.
  - Both solutions contain freely moving molecules or particles.
- Reasoning
  - $\because$ An electric field exerts opposite forces on charged particles but nearly none on neutral molecules.
  - $\therefore$ Only mobile *ions* transport charge through the solution.
- Takeaway
  - Mobile charge carriers determine electrical conduction.

Why Does Conductivity Vary Between Ionic Solutions?

- Question
  - Why do equally concentrated electrolytes produce different conductivities?
- Evidence
  - Solutions with more dissolved ions usually conduct more strongly.
  - Different ions produce different conductivities at equal concentration.
- Reasoning
  - $\because$ Current increases when more ions drift, and faster-moving ions carry charge more efficiently.
  - $\therefore$ Conductivity reflects both *ion concentration* and ion mobility.
- Takeaway
  - Conductivity measures available moving charge.

What Does the Equilibrium Arrow Mean?

- Question
  - What microscopic process does the symbol $\rightleftharpoons$ represent?
- Evidence
  - Weak electrolytes always contain both molecules and ions.
  - Conductivity remains constant after the solution settles.
- Reasoning
  - $\because$ Molecules continuously ionize while ions simultaneously recombine into molecules.
  - $\therefore$ Constant composition reflects equal forward and reverse reaction rates.
- Takeaway
  - Equilibrium is dynamic, not motionless.

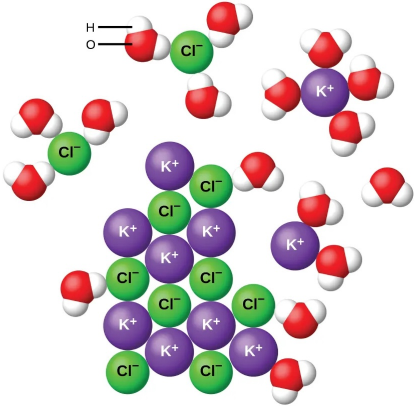
Why Do Strong and Weak Electrolytes Differ?

- Question
  - Why does one electrolyte ionize almost completely while another does not?
- Evidence
  - Strong electrolytes produce far greater conductivity than weak electrolytes.
  - Weak electrolytes retain many neutral molecules at equilibrium.
- Reasoning
  - $\because$ Molecular structure determines whether separated ions or intact molecules have lower overall energy in water.
  - $\therefore$ The equilibrium favors whichever state is energetically more stable.
- Takeaway
  - Structure determines equilibrium ionization.

Ionic Electrolytes: Dissociation (Physical Change)

For **ionic compounds** (e.g., KCl):

1. Water molecules cluster around surface ions.
2. **Ion-dipole forces** (between the ion and polar water) overcome the electrostatic lattice forces.
3. Ions separate and disperse into solution — a **physical change**.



O-end ($\delta-$) of water orients toward **cations**;
H-end ($\delta+$) orients toward **anions**.

As potassium chloride (KCl) dissolves in water, the ions are hydrated. The polar water molecules are attracted by the charges on the K^+ and Cl^- ions. Water molecules in front of and behind the ions are not shown.

Covalent Electrolytes: Ionization (Chemical Change)

Covalent compounds become electrolytes by **chemically reacting** with water to produce ions — a **chemical change** (ionization).

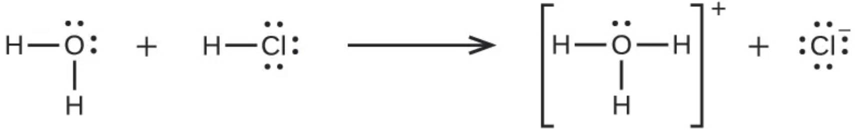
Key distinction: ionic electrolytes undergo **physical** dissociation; covalent electrolytes undergo **chemical** ionization.

Hydrogen Chloride (HCl) — Strong Acid

Pure HCl gas: covalent molecules, no ions, no conductivity.

Dissolved in water: $HCl(g) + H_2O(\ell) \longrightarrow H_3O^+(aq) + Cl^-(aq)$

Reaction is $\sim$100% complete $\Rightarrow$ **strong electrolyte**.



strong acids are *strong electrolytes* because they readily dissociate into ions.

weak acids are *weak electrolytes* because they weakly dissociate into ions.

Covalent Electrolytes: Ionization (Chemical Change)

Water — Extremely Weak Electrolyte

Only ~2 out of every **1 billion** water molecules ionize at 25 °C:

$$H_2O(\ell) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$$

This is why pure water is an extremely poor conductor of electricity; it is only very slightly ionized.

Why Can Water Separate Ions and Accept Protons?

- Question
  - What structural feature enables water to perform these two different roles?
- Evidence
  - Ionic compounds dissolve into ions in water.
  - Acids transfer protons to water, forming H_3O^+.
- Reasoning
  - $\because$ Water has polar O–H bonds and oxygen *lone pairs*, creating uneven electron density.
  - $\therefore$ Water stabilizes ions electrostatically and donates lone-pair electrons to bind incoming protons.
- Takeaway
  - Electron distribution determines water's chemical roles.

Why Can Water Pull Apart an Ionic Crystal?

- Question
  - Why can water overcome the attractions holding an ionic lattice together?
- Evidence
  - Many ionic solids dissolve readily in water.
  - Hydration releases hundreds of kJ mol^{-1} per ion.
- Reasoning
  - $\because$ Polar water molecules surround separated ions, creating many strong ion–dipole attractions that lower potential energy.
  - $\therefore$ Hydration can compensate for lattice attraction, allowing *dissociation* into existing ions.
- Takeaway
  - Strong solvation stabilizes separated ions.

Why Do Some Covalent Acids Ionize Completely?

- Question
  - Why does HCl ionize almost completely while acetic acid usually does not?
- Evidence
  - HCl forms nearly all H_3O^+ and Cl^- in water.
  - Acetic acid remains mostly neutral molecules.
- Reasoning
  - $\because$ Molecular electron distribution determines how easily proton transfer lowers the system's potential energy.
  - $\therefore$ HCl favors *ionization*, whereas acetic acid usually favors retaining its proton.
- Takeaway
  - Molecular structure sets proton-transfer equilibrium.

Why Does Pure HCl Not Conduct Electricity?

- Question
  - Why does liquid HCl conduct poorly until water is added?
- Evidence
  - Pure liquid HCl has negligible conductivity.
  - Aqueous HCl conducts strongly.
- Reasoning
  - $\because$ Pure HCl lacks a species that efficiently stabilizes separated charges after proton transfer.
  - $\therefore$ Water creates stable mobile ions, enabling sustained electrical conduction.
- Takeaway
  - Stable ions require a stabilizing solvent.

Check Your Understanding

Classify each as a **strong electrolyte**, **weak electrolyte**, or **nonelectrolyte**:

(a) NaCl (table salt)

(b) Ethanol (C_2H_5OH)

(c) Acetic acid (CH_3COOH)

(d) HCl (hydrochloric acid)

Come to lecture to see the answer.

Part III

Solubility

Outline of Part III: Solubility

Factors Affecting Solubility

Outline of Part III: Solubility

Factors Affecting Solubility

Solubility: Key Terms

Definition: Solubility

The **solubility** of a substance is its maximum concentration in solution at equilibrium, under specified conditions of temperature and pressure.

Concentration Vocabulary

- **Saturated**: at the solubility limit (equilibrium between dissolving and crystallizing).
- **Unsaturated**: below the limit — more solute can dissolve.
- **Supersaturated**: above the limit — unstable; crystallization can be triggered by mechanical shock or seeding.

Other Useful Terms

- **Dilute**: relatively low solute concentration.
- **Concentrated**: relatively high solute concentration.
- **Miscible**: liquids that mix in all proportions.
- **Immiscible**: liquids with minimal mutual solubility.

Why Is a Saturated Solution an Equilibrium?

- Question
  - Why does a saturated solution appear unchanged although particles still move?
- Evidence
  - Undissolved solid remains while solution concentration stays constant.
  - Individual ions continually exchange between crystal and solution.
- Reasoning
  - $\because$ Thermal motion continuously causes both dissolution and crystal attachment at the solid surface.
  - $\therefore$ Equal opposing rates produce constant concentration despite ongoing microscopic exchange.
- Takeaway
  - Equilibrium balances opposing microscopic processes.

Why Can Supersaturated Solutions Persist?

- Question
  - Why does excess dissolved solute sometimes remain instead of crystallizing immediately?
- Evidence
  - Supersaturated solutions can remain clear for extended periods.
  - Seeding or mechanical shock often triggers rapid crystallization.
- Reasoning
  - $\because$ Forming the first stable crystal requires organizing many particles into a sufficiently large nucleus.
  - $\therefore$ Disturbance overcomes this kinetic barrier, allowing crystallization toward equilibrium.
- Takeaway
  - Kinetics can delay thermodynamically favored change.

Why Does Changing Conditions Shift Solubility?

- Question
  - Why do temperature or pressure change the amount that remains dissolved?
- Evidence
  - Solubility changes when temperature or pressure changes.
  - The final dissolved concentration reaches a new constant value.
- Reasoning
  - $\because$ External conditions alter the relative stability of dissolved and crystalline states, changing opposing dissolution and crystallization rates.
  - $\therefore$ The equilibrium shifts until equal rates are reestablished at a different concentration.
- Takeaway
  - Equilibrium responds to changes in state stability.

"Like Dissolves Like"

The General Rule

Substances with **similar intermolecular forces** tend to dissolve in each other:

- **Polar solvents** (e.g., water) dissolve **polar and ionic** solutes.
- **Nonpolar solvents** (e.g., hexane, CCl_4) dissolve **nonpolar** solutes.

Miscible with Water

Ethanol, sulfuric acid, ethylene glycol — all polar; can form hydrogen bonds or strong dipole interactions with water.

Immiscible with Water

Gasoline, benzene, CCl_4 — all nonpolar; weak dispersion forces cannot compete with water's strong hydrogen bonds.

Oil and water don't mix because oil is nonpolar and water's strong H-bonds with itself exclude nonpolar molecules.

Why Does the Balance of Intermolecular Forces Control Mixing?

- Question
  - Why does miscibility depend on three interaction strengths instead of one?
- Evidence
  - Mixing breaks solute–solute and solvent–solvent attractions.
  - Mixing simultaneously forms new solute–solvent attractions.
- Reasoning
  - $\because$ Molecules rearrange only if new attractions replace those broken without greatly increasing potential energy.
  - $\therefore$ Mixing is favored when *solute–solvent* attractions approximately match the original interactions.
- Takeaway
  - Mixing compares interactions before and after rearrangement.

Why Does Water Exclude Nonpolar Molecules?

- Question
  - Why do nonpolar solutes dissolve poorly although mixing increases possible arrangements?
- Evidence
  - Oil separates from water after brief mixing.
  - Water molecules become highly ordered around isolated nonpolar solutes.
- Reasoning
  - $\because$ Water preserves its hydrogen-bond network by forming ordered shells around nonpolar surfaces, reducing molecular freedom.
  - $\therefore$ Aggregating nonpolar molecules releases ordered water, making phase separation more favorable.
- Takeaway
  - Water minimizes disruption of its hydrogen-bond network.

Why Is Hydrogen Bonding Especially Effective?

- Question
  - Why does hydrogen bonding promote miscibility more effectively than ordinary dipole interactions?
- Evidence
  - Methanol mixes completely with water.
  - Longer alcohols become progressively less miscible.
- Reasoning
  - $\because$ *Hydrogen bonds* are highly directional, allowing molecules to replace water–water bonds with similarly strong new connections.
  - $\therefore$ Small hydrogen-bonding molecules mix readily until nonpolar regions dominate their interactions.
- Takeaway
  - Directional bonding can offset disrupted intermolecular attractions.

Gas Solubility: Effect of Temperature

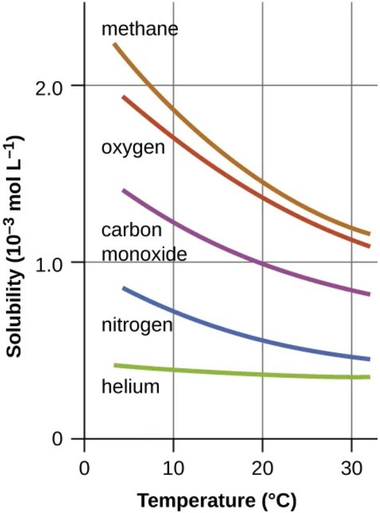
Gas solubility in liquids **decreases** as temperature increases.

As temperature rises, dissolved gas molecules gain kinetic energy to escape the solution — the equilibrium shifts toward the gas phase.

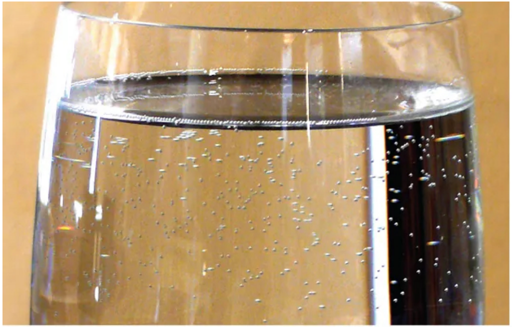
Environmental Consequence: Thermal Pollution

Warm-water discharge from power plants reduces dissolved O_2.

Freshwater trout require $\geq 7.5\ \frac{mg}{L}$ dissolved O_2 to survive — warm streams may fall below this threshold.



Temperature Reduces Gas Solubility in Water

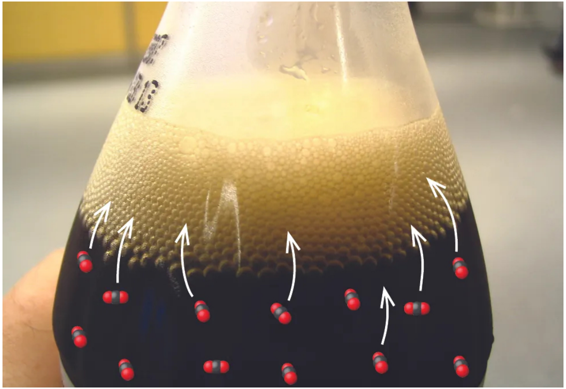


(a)

(b)

As water temperature increases, the solubility of gases decreases. In a glass of water, dissolved air escapes as visible bubbles as the water warms. In lakes and rivers, reduced dissolved oxygen can stress or kill aquatic organisms, making thermal pollution a serious environmental concern.

Gas Solubility Increases with Pressure



Carbonated beverages are bottled under high carbon dioxide (CO_2) pressure, allowing more CO_2 to dissolve in the liquid. When the bottle is opened, the pressure drops, decreasing CO_2 solubility. Excess dissolved CO_2 escapes as bubbles until a new equilibrium is reached, eventually causing the beverage to go flat.

Gas Solubility: Henry's Law

Henry's Law

The concentration of a dissolved gas is **directly proportional** to its partial pressure above the solution:

$$\boxed{C_g = k \cdot P_g}$$

C_g = dissolved gas concentration $\left(\frac{\text{mol}}{\text{L}}\right)$, P_g = partial pressure of the gas, k = Henry's Law constant (gas-, solvent-, and temperature-specific).

- **Carbonated drinks**: bottled under high CO_2 pressure; opening releases pressure $\Rightarrow$ CO_2 solubility drops $\Rightarrow$ bubbles.
- **The "bends" (decompression sickness)**: N_2 dissolves into blood under high pressure during deep dives; rapid ascent causes N_2 bubbles to form in tissues.
- **Lake Nyos (1986), Cameroon**: volcanic CO_2 had saturated the deep lake under high pressure; sudden lake overturn released a massive CO_2 cloud that killed ~1700 people.

Example: Applying Henry's Law

Henry's Law Calculation — (chem2e_ch11_ex11.2)

At 20 °C, the Henry's Law constant for O_2 in water is $k = 1.36 \times 10^{-5} \, \frac{\text{mol}}{\text{L kPa}}$.

(a) What is $[O_2]$ when $P_{O_2} = 101.3 \, \text{kPa}$ (pure O_2)?
(b) What is $[O_2]$ at atmospheric conditions ($P_{O_2} = 20.7 \, \text{kPa}$)?

(a) Pure O_2:

$$C_g = k \cdot P_g$$

$$= 1.36 \times 10^{-5} \, \frac{\text{mol}}{\text{L kPa}} \times 101.3 \, \text{kPa}$$

$$\boxed{C_g = 1.38 \times 10^{-3} \, \frac{\text{mol}}{\text{L}}}$$

(b) Atmospheric O_2:

$$C_g = 1.36 \times 10^{-5} \, \frac{\text{mol}}{\text{L kPa}} \times 20.7 \, \text{kPa}$$

$$\boxed{C_g = 2.82 \times 10^{-4} \, \frac{\text{mol}}{\text{L}}}$$

Pressure Changes, Henry's Law, and Decompression Sickness

(a)

(b)

(a) US Navy divers undergo training in a recompression chamber. (b) Divers receive hyperbaric oxygen therapy.

As a diver descends, increased pressure causes more gases, especially nitrogen, to dissolve in the blood and body fluids. If the diver ascends too quickly, the sudden pressure drop reduces gas solubility, allowing dissolved gases to form bubbles that can cause decompression sickness ("the bends"). Recompression chambers restore high pressure and deliver oxygen to help dissolve the bubbles and safely eliminate excess gas from the body.

Why Does a Gas Dissolve in a Liquid?

- Question
  - Why can gas molecules leave the gas phase and remain dissolved?
- Evidence
  - Many gases dissolve spontaneously in water.
  - Gas dissolution usually releases measurable heat.
- Reasoning
  - $\because$ Solvent molecules form attractive interactions that lower a dissolved gas molecule's potential energy despite restricting its motion.
  - $\therefore$ Dissolution is an exothermic equilibrium between energetic stabilization and reduced molecular freedom.
- Takeaway
  - Gas solubility balances energy and molecular freedom.

Why Do Temperature and Pressure Shift Gas Solubility?

- Question
  - Why do heating and compression change dissolved gas concentration?
- Evidence
  - Heating usually decreases gas solubility.
  - Increasing gas pressure increases dissolved gas concentration.
- Reasoning
  - $\because$ Heating favors higher-energy gas molecules, whereas compression increases collisions driving molecules into solution.
  - $\therefore$ The equilibrium gas(g) $\rightleftharpoons$ gas(aq) shifts opposite directions for temperature and pressure changes.
- Takeaway
  - External conditions shift dissolution equilibrium.

Why Is Henry's Law Linear?

- Question
  - Why does dissolved gas concentration increase proportionally with pressure?
- Evidence
  - Dilute gases approximately obey $C = kP$.
  - The proportionality fails at sufficiently high pressures.
- Reasoning
  - $\because$ At low concentration, dissolved gas molecules rarely interact, so each responds independently to increased gas-phase collisions.
  - $\therefore$ Each pressure increase adds nearly the same dissolved amount, producing an approximately linear relationship.
- Takeaway
  - Independent particles produce linear equilibrium behavior.

When Does Henry's Law Fail?

- Question
  - Why do some gases deviate from Henry's Law?
- Evidence
  - CO_2 and NH_3 often exceed simple Henry's Law predictions.
  - Deviations increase at high pressure.
- Reasoning
  - $\because$ Chemical reactions or strong gas–gas interactions create additional stabilization absent from the dilute independent-particle picture.
  - $\therefore$ Concentration no longer changes proportionally with pressure.
- Takeaway
  - Linear laws require weakly interacting dissolved particles.

Why Do Different Gases Respond Differently to Temperature?

- Question
  - Why does helium show weaker temperature dependence than methane?
- Evidence
  - Helium solubility changes less with temperature than methane.
  - Methane interacts more strongly with surrounding molecules.
- Reasoning
  - $\because$ Stronger *intermolecular attractions* release more energy during dissolution, making heating oppose dissolution more strongly.
  - $\therefore$ Gases with stronger solvent attractions exhibit greater temperature sensitivity.
- Takeaway
  - Interaction strength controls thermal response.

Solutions of Liquids in Liquids: Miscibile Liquids

Miscible liquids mix in all proportions to form a single homogeneous solution.

- Water and ethanol
- Water and sulfuric acid
- Water and ethylene glycol (antifreeze)
- Gasoline and two-cycle motor oil

Miscibility depends primarily on intermolecular forces.

Like dissolves like:

- Polar liquids dissolve in polar liquids.
- Nonpolar liquids dissolve in nonpolar liquids.

Similar intermolecular attractions between solute and solvent favor mixing.

Solutions of Liquids in Liquids: Immiscible Liquids

Immiscible liquids have very low mutual solubility and form separate layers.

- Water and oil
- Water and gasoline
- Water and benzene
- Water and carbon tetrachloride

Water molecules are held together by strong hydrogen bonding.

Why don't they mix?

Weak attractions between water and nonpolar molecules cannot overcome the strong water–water interactions.

The less dense liquid forms the upper layer (e.g., oil floats on water).

Solutions of Liquids in Liquids: Partially Miscible Liquids

Partially miscible liquids dissolve in each other only to a limited extent.

Example: bromine and water.

After Mixing

- Upper layer: water saturated with bromine.
- Lower layer: bromine saturated with water.

Partial miscibility is an intermediate case between complete miscibility and immiscibility. Each layer contains a small amount of the other liquid at equilibrium.

Why Do Some Liquids Mix Completely While Others Do Not?

- Question
  - What determines whether two liquids form one phase or two?
- Evidence
  - Mixing breaks liquid–liquid attractions in both pure liquids.
  - Mixing simultaneously forms new liquid–liquid attractions between unlike molecules.
- Reasoning
  - $\because$ Mixing lowers free energy only when new *solute–solvent* attractions compensate for disrupted self-attractions.
  - $\therefore$ Miscibility depends on the energetic balance before and after molecular rearrangement.
- Takeaway
  - Mixing requires energetically favorable molecular rearrangement.

Why Are Oil and Water Immiscible?

- Question
  - Why does entropy alone not keep oil mixed with water?
- Evidence
  - Oil and water separate after standing.
  - Water molecules become ordered around isolated nonpolar molecules.
- Reasoning
  - $\because$ Water preserves its cooperative *hydrogen-bond* network by forming ordered hydration shells around nonpolar surfaces.
  - $\therefore$ Aggregating nonpolar molecules releases ordered water, lowering the mixture's free energy.
- Takeaway
  - Water minimizes disruption of its hydrogen-bond network.

Why Is Hydrogen Bonding Especially Effective?

- Question
  - Why does hydrogen bonding promote miscibility better than ordinary dipole interactions?
- Evidence
  - Methanol mixes completely with water.
  - Longer alcohols become progressively less miscible.
- Reasoning
  - $\because$ Hydrogen bonds are strong and directional, allowing replacement of water–water bonds with similarly stable new connections.
  - $\therefore$ Small hydrogen-bonding molecules remain miscible until nonpolar regions dominate their interactions.
- Takeaway
  - Directional interactions stabilize molecular mixing.

Why Are Some Liquids Only Partially Miscible?

- Question
  - Why do some liquid pairs separate into two equilibrium liquid phases?
- Evidence
  - Each layer contains measurable amounts of the other liquid.
  - The compositions remain constant after separation.
- Reasoning
  - $\because$ Limited mutual dissolution lowers free energy, but complete mixing raises it through unfavorable intermolecular interactions.
  - $\therefore$ Two saturated liquid phases become the stable equilibrium state.
- Takeaway
  - Equilibrium can consist of two solution phases.

Why Does Density Only Determine Layer Order?

- Question
  - Why does density affect layer position but not miscibility?
- Evidence
  - Immiscible liquids form separate layers.
  - Either liquid may occupy the upper layer.
- Reasoning
  - $\because$ Intermolecular interactions determine whether phases separate, whereas gravity only arranges already separated phases.
  - $\therefore$ Density determines layer order after phase separation has occurred.
- Takeaway
  - Stability precedes gravitational arrangement.

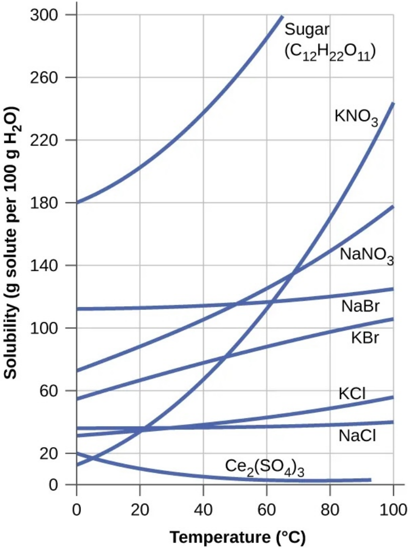
Solid Solubility: Effect of Temperature

For *nearly all* **solids**, solubility **increases** with temperature.

- **KNO_3**: dramatically more soluble at 80 °C than 20 °C.
- **NaCl**: nearly constant over 0–100 °C.
- **$Ce_2(SO_4)_3$**: *decreases* with temperature (notable exception).

Application: Hand Warmers

Sodium acetate ($NaC_2H_3O_2$) forms a **supersaturated** solution when cooled. Clicking a metal disc triggers rapid **exothermic crystallization**, releasing heat.



Temperature Reveals the Driving Force

- Question
  - Why do some solids become more soluble when heated while others become less soluble?
- Evidence
  - Most salts dissolve better at higher temperature.
  - Some (notably $Ce_2(SO_4)_3$) become less soluble as temperature rises.
- Reasoning
  - $\because$ Dissolution balances lattice disruption against ion hydration, releasing or absorbing heat.
  - $\therefore$ Temperature favors whichever equilibrium direction consumes the added thermal energy.
- Takeaway
  - Heat shifts equilibria according to dissolution enthalpy.

Why Different Solids Respond Differently

- Question
  - What molecular property determines each solid's temperature dependence?
- Evidence
  - NaCl solubility changes little with temperature.
  - Many nitrates show much larger increases.
- Reasoning
  - $\because$ Different ions change lattice and hydration energies by different amounts during dissolution.
  - $\therefore$ The net dissolution enthalpy sets both direction and sensitivity to temperature.
- Takeaway
  - Competing interaction energies determine thermal response.

Quantifying the Equilibrium Shift

- Question
  - How does equilibrium quantitatively connect temperature and solubility?
- Evidence
  - Measured solubilities correspond to equilibrium constants.
  - $d(\ln K)/dT = \Delta H^\circ/(RT^2)$ matches experimental trends.
- Reasoning
  - $\because$ ΔH° measures heat exchanged as equilibrium advances by one reaction amount.
  - $\therefore$ Its sign determines whether increasing temperature raises or lowers the equilibrium constant.
- Takeaway
  - Enthalpy predicts equilibrium sensitivity to temperature.

One Principle for Every Solubility Equilibrium

- Question
  - Why do gases, solids, and later equilibria follow the same temperature rule?
- Evidence
  - Gas solubility usually decreases with heating.
  - Solid solubility may increase or decrease with heating.
- Reasoning
  - $\because$ Every dissolution process is an equilibrium with its own enthalpy change.
  - $\therefore$ A single equilibrium framework replaces separate memorized solubility rules.
- Takeaway
  - Equilibrium enthalpy unifies temperature-dependent solubility.

Supersaturated Solutions and Heat Packs

Reusable hand warmers contain a supersaturated solution of sodium acetate and a flexible metal disk. Clicking the disk initiates crystallization, causing the excess dissolved sodium acetate to rapidly precipitate. This crystallization process releases heat, warming the pack to about 130°F (54°C). After use, the solid sodium acetate can be dissolved again by heating the pack in boiling water, restoring the supersaturated solution for reuse.

Check Your Understanding

A warm can of soda is opened and fizzes violently.

Explain this using **two** concepts from this section.

(Hint: think about both temperature and pressure.)

Come to lecture to see the answer.

Part IV

Colligative Properties

Outline of Part IV: Colligative Properties

Vapor Pressure Lowering

B.P. Elevation and F.P. Depression

Osmosis and Osmotic Pressure

Colligative Properties: Overview

Definition: Colligative Properties

Properties of solutions that depend only on the **total number** of dissolved solute particles, **not** on the chemical identity of those particles.

The Four Colligative Properties

1. **Vapor pressure lowering**
2. **Boiling point elevation**
3. **Freezing point depression**
4. **Osmotic pressure**

1 mol of glucose, 1 mol of NaCl (which gives 2 mol of ions), and 1 mol of urea are chemically different — but a given *number of particles* affects colligative properties the same way, regardless of identity.

One Hidden Variable Behind Four Properties

- Question
  - Why do four different solution properties depend only on dissolved particle number?
- Evidence
  - Dilute nonelectrolytes show identical effects at equal particle concentration.
  - Different solute identities give nearly identical changes when particle counts match.
- Reasoning
  - $\because$ Dilute solutes mainly reduce the fraction of solvent molecules, while solute–solute interactions remain negligible.
  - $\therefore$ The solvent responds primarily to *mole fraction*, not chemical identity.
- Takeaway
  - Particle number controls dilute-solution thermodynamics.

Why Solvent Chemical Potential Falls

- Question
  - Why does adding particles lower the solvent's chemical potential?
- Evidence
  - Mixing increases the number of accessible molecular arrangements.
  - The solvent occupies a smaller fraction of all molecules after mixing.
- Reasoning
  - $\because$ Greater configurational freedom makes each solvent molecule statistically less likely to escape unchanged.
  - $\therefore$ The solvent reaches a lower *chemical potential*, determined by its mole fraction.
- Takeaway
  - Mixing lowers solvent escaping tendency.

Four Consequences of One Potential Change

- Question
  - How can one lowered solvent potential produce four different observable properties?
- Evidence
  - Vapor pressure decreases; boiling and freezing temperatures shift; osmotic pressure appears.
  - All effects increase as dissolved particle number increases.
- Reasoning
  - ∵ Every phase equilibrium compares the solvent's chemical potential between competing states.
  - ∴ One lowered solvent potential shifts every equilibrium consistently.
- Takeaway
  - One thermodynamic change explains four properties.

Why the Dilute Approximation Eventually Fails

- Question
  - Why do concentrated solutions no longer obey ideal colligative behavior?
- Evidence
  - Measured colligative properties increasingly deviate at high concentrations.
  - Electrolytes show especially large departures from ideal predictions.
- Reasoning
  - $\because$ Solute particles increasingly interact, reducing their independent contribution to solvent statistics.
  - $\therefore$ Mole fraction alone no longer completely determines solvent chemical potential.
- Takeaway
  - Particle interactions limit dilute-solution predictions.

Concentration Units: Mole Fraction and Molality

Mole Fraction (X)

$$X_A = \frac{n_A}{n_{\text{total}}}$$

n_A = moles of component A
n_{total} = total moles of all components
Dimensionless; $\sum_i X_i = 1$.

Molality (m)

$$m = \frac{n_{\text{solute}}}{m_{\text{solvent}}\,(\text{kg})}$$

Units: $\frac{\text{mol}}{\text{kg}}$

Temperature-independent: mass does not change with T, unlike volume (molarity).

Both mole fraction and molality are **temperature-independent** — preferred for colligative property calculations where temperature changes substantially.

Why Molality Tracks Colligative Physics

- Question
  - Why do freezing- and boiling-point equations use *molality* instead of molarity?
- Evidence
  - Heating changes solution volume but scarcely changes solvent mass.
  - Colligative properties vary reproducibly with temperature.
- Reasoning
  - $\because$ Solvent chemical potential depends on particle ratio, requiring composition unchanged while temperature varies.
  - $\therefore$ Mass-based concentration isolates particle ratio; volume-based concentration introduces unrelated thermal expansion.
- Takeaway
  - Use temperature-invariant composition to predict equilibrium shifts.

Why Molality Approximates Mole Fraction

- Question
  - Why can *molality* replace solvent mole fraction in dilute solutions?
- Evidence
  - Dilute solutions contain far fewer solute particles than solvent molecules.
  - Measured colligative effects increase nearly linearly with molality.
- Reasoning
  - $\because$ Adding few solute particles scarcely changes solvent amount, making particle ratio nearly proportional to molality.
  - $\therefore$ The proportionality becomes embedded within the *cryoscopic* and ebullioscopic constants.
- Takeaway
  - Dilute approximations convert mole fraction into molality.

When Molality-Based Equations Fail

- Question
  - Why do colligative-property equations lose accuracy in concentrated solutions?
- Evidence
  - Concentrated solutions deviate systematically from linear predictions.
  - Solute particles interact more frequently as concentration increases.
- Reasoning
  - $\because$ Particle interactions change solvent environments, so particle count alone no longer determines solvent chemical potential.
  - $\therefore$ Molality no longer uniquely predicts equilibrium shifts without nonideal corrections.
- Takeaway
  - Particle count governs only the dilute-limit behavior.

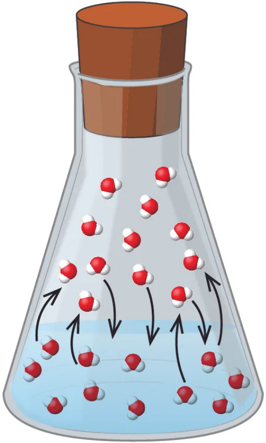
Outline of Part IV: Colligative Properties

Vapor Pressure Lowering

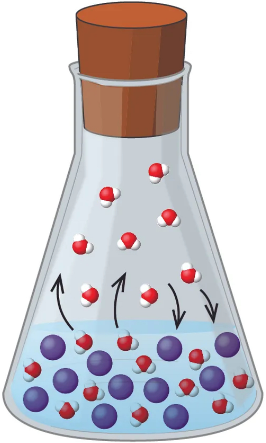
B.P. Elevation and F.P. Depression

Osmosis and Osmotic Pressure

Vapor Pressure Lowering by Nonvolatile Solutes



(a) Pure water



(b) Aqueous solution

A pure solvent reaches equilibrium when evaporation and condensation occur at equal rates. When a nonvolatile solute is dissolved in the solvent, fewer solvent molecules are present at the liquid surface and able to escape into the vapor phase. As a result, the rate of evaporation decreases, equilibrium is established at a lower vapor pressure, and the solution has a lower vapor pressure than the pure solvent.

Vapor Pressure Lowering: Raoult's Law

Raoult's Law

The partial pressure of a volatile component above its solution equals the product of its **mole fraction** and its **pure-component vapor pressure**:

$$P_A = X_A \cdot P_A^*$$

For a solution with a **nonvolatile solute**:

$$P_{\text{soln}} = X_{\text{solvent}} \cdot P_{\text{solvent}}^*$$

Since $X_{\text{solvent}} < 1$, the vapor pressure above a solution is always **lower** than above the pure solvent.

Physical basis: Nonvolatile solute molecules occupy surface sites, reducing the evaporation rate of solvent.

Why Vapor Pressure Falls After Dissolving a Solute

- Question
  - Why does dissolved solute reduce vapor pressure beyond surface blocking?
- Evidence
  - Dilute KCl lowers vapor pressure despite negligible surface coverage.
  - Molecular and ionic solutes produce similar trends at equal solvent fraction.
- Reasoning
  - $\because$ Mixing increases accessible molecular arrangements, stabilizing the liquid by lowering solvent *chemical potential*.
  - $\therefore$ Fewer solvent molecules escape before liquid and vapor reach equilibrium.
- Takeaway
  - Phase equilibrium depends on bulk molecular stability.

Deriving Raoult's Law from Chemical Potential

- Question
  - How does reduced liquid chemical potential determine vapor pressure?
- Evidence
  - Equilibrium requires equal solvent chemical potentials in liquid and vapor.
  - For an ideal solution, $\mu_\ell = \mu^* + RT \ln x_{\text{solvent}}$.
- Reasoning
  - $\because$ Lower liquid chemical potential must be matched by lower vapor chemical potential, achieved through fewer vapor molecules.
  - $\therefore$ Equality gives $P = x_{\text{solvent}} P^*$, where x measures solvent fraction and P^* is pure-solvent vapor pressure.
- Takeaway
  - Equilibrium converts molecular stability into measurable pressure.

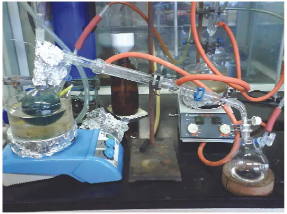
Why Raoult's Law Is Linear

- Question
  - Why does vapor pressure scale linearly with solvent fraction only sometimes?
- Evidence
  - Many dilute mixtures follow $P = xP^*$.
  - Some mixtures show positive or negative deviations from this line.
- Reasoning
  - $\because$ Linear behavior requires replacing neighbors without changing average intermolecular attraction in the liquid.
  - $\therefore$ Stronger or weaker new attractions alter liquid stabilization, shifting vapor pressure above or below Raoult's prediction.
- Takeaway
  - Intermolecular forces control deviations from linearity.

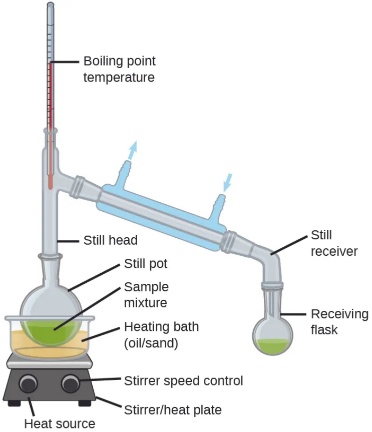
Meaning of an Ideal Solution

- Question
  - Why is the ideal-solution assumption broadly successful?
- Evidence
  - Many similar liquids obey Raoult's Law over substantial composition ranges.
  - Real gases depart from ideality whenever intermolecular attractions become important.
- Reasoning
  - $\because$ Ideal solutions require similar interaction energies, not zero attractions, so mixing barely changes liquid stabilization.
  - $\therefore$ The same chemical-potential argument extends directly to freezing-point depression and other phase equilibria.
- Takeaway
  - Similar interactions preserve transferable equilibrium reasoning.

Laboratory Distillation: Separating Volatile Liquids by Differences in Boiling Behavior



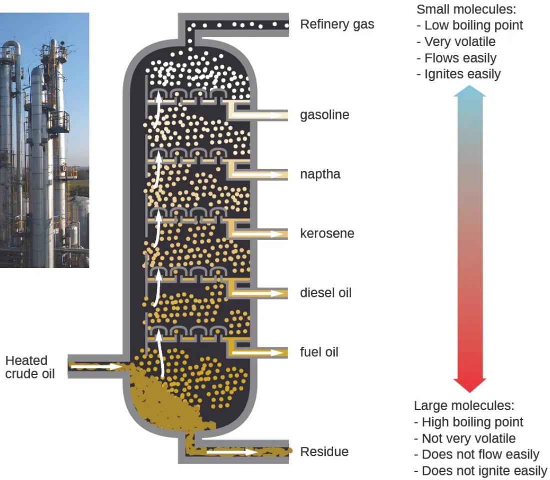
(a)



(b)

A typical laboratory distillation apparatus, shown as both a photograph and a labeled schematic, illustrates how mixtures of volatile liquids are separated. Heating vaporizes the more volatile component first, producing a vapor enriched in that component, which is then condensed and collected. This process relies on differences in vapor pressure (described by Raoult's law) and corresponding boiling behavior to achieve separation and purification.

Fractional Distillation of Crude Oil: Separating Hydrocarbons by Boiling Point



This figure shows how a refinery uses a fractional distillation column to separate crude oil into useful petroleum fractions. As heated crude oil rises through the column, hydrocarbons condense at different heights according to their boiling points and volatility—larger, higher-boiling molecules condense near the bottom, while smaller, more volatile molecules rise higher before condensing. This continuous process produces products such as fuel oil, diesel, kerosene, naphtha, gasoline, and refinery gases.

Outline of Part IV: Colligative Properties

Vapor Pressure Lowering

B.P. Elevation and F.P. Depression

Osmosis and Osmotic Pressure

Boiling Point Elevation and Freezing Point Depression

Boiling Point Elevation

The lower vapor pressure of the solution means a **higher temperature** is needed to reach atmospheric pressure and boil:

$$\Delta T_b = K_b \cdot m$$

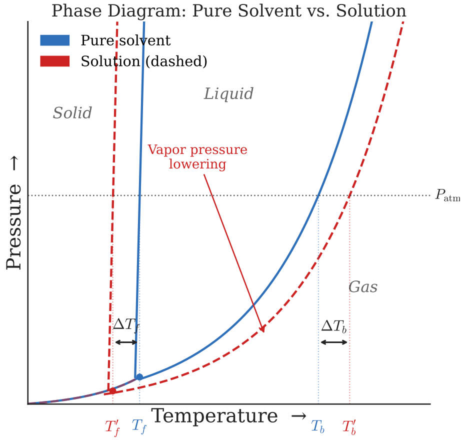
$\Delta T_b =$ boiling point elevation (°C)
$K_b =$ ebullioscopic constant $\left(\frac{°\mathrm{C\,kg}}{\mathrm{mol}}\right)$
$m =$ molality $\left(\frac{\mathrm{mol}}{\mathrm{kg}}\right)$

Freezing Point Depression

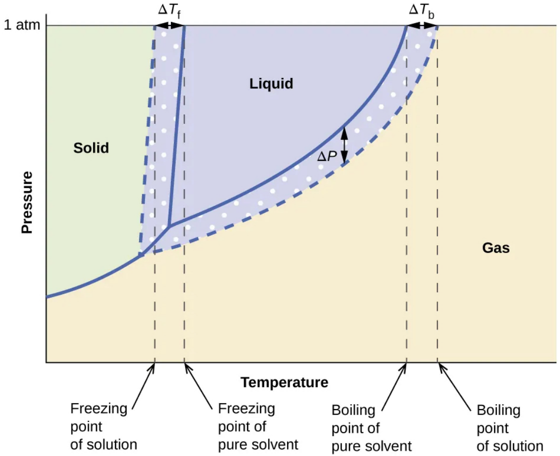
A solute lowers the freezing point:

$$\Delta T_f = K_f \cdot m$$

$K_f =$ cryoscopic constant $\left(\frac{°\mathrm{C\,kg}}{\mathrm{mol}}\right)$



Phase Diagram for Pure Solvent v. Solution



Phase diagrams for a pure solvent (solid curves) and a solution formed by dissolving nonvolatile solute in the solvent (dashed curves).

Constants and Applications

Water Constants

$$K_b = 0.512\ \frac{°\mathrm{C\,kg}}{\mathrm{mol}}$$
$$K_f = 1.86\ \frac{°\mathrm{C\,kg}}{\mathrm{mol}}$$
$$\text{Normal BP} = 100\,°\mathrm{C}$$
$$\text{Normal FP} = 0\,°\mathrm{C}$$

Real-World Applications

- **Road de-icing**: NaCl and $CaCl_2$ lower the freezing point of water on roads.
- **Antifreeze**: Ethylene glycol in car radiators prevents both freezing and boiling.
- **Cooking**: Adding salt to pasta water raises the boiling point slightly (though minimally in practice).

Example: BP Elevation and FP Depression

Ethylene Glycol

62.1 g of ethylene glycol ($C_2H_6O_2$, $M = 62.07\,\frac{\text{g}}{\text{mol}}$) is dissolved in 250 g of water. Find the boiling point elevation and freezing point depression.

Moles of solute:

$$n = \frac{62.1\,\text{g}}{62.07\,\frac{\text{g}}{\text{mol}}} = 1.00\,\text{mol}$$

Molality:

$$m = \frac{1.00\,\text{mol}}{0.250\,\text{kg}} = 4.00\,\frac{\text{mol}}{\text{kg}}$$

Boiling point elevation:

$$\boxed{\Delta T_b = 0.512 \times 4.00 = 2.05\,^\circ\text{C}}$$

Freezing point depression:

$$\boxed{\Delta T_f = 1.86 \times 4.00 = 7.44\,^\circ\text{C}}$$

New BP: $102.05\,^\circ\text{C}$

New FP: $-7.44\,^\circ\text{C}$

Why Ice Rejects Dissolved Solute

- Question
  - Why does freezing produce nearly pure ice instead of trapping dissolved ions?
- Evidence
  - Seawater forms almost pure ice during freezing.
  - Salt concentration increases in the remaining liquid.
- Reasoning
  - $\because$ Ice is stabilized by a regular hydrogen-bond network that ions disrupt, raising the solid's potential energy.
  - $\therefore$ The solid remains nearly pure, while solute stays in the liquid and lowers its *chemical potential*.
- Takeaway
  - Stable crystal structures exclude incompatible solutes.

Why Freezing Point Depression Depends Only on the Solvent

- Question
  - How does lowered liquid chemical potential determine the freezing temperature?
- Evidence
  - Freezing begins when solvent chemical potentials in solid and liquid are equal.
  - Experiments give $\Delta T_f = K_f m$ for dilute ideal solutions.
- Reasoning
  - $\because$ Cooling reduces the liquid chemical potential until it again matches the unchanged pure-solid value.
  - $\therefore$ Clausius–Clapeyron converts that required potential change into $K_f = \dfrac{RT_{fp}^2 M}{\Delta H_{fus}}$, where only solvent properties appear.
- Takeaway
  - Solvent energetics determine freezing-point sensitivity.

Why the Phase Boundaries Shift in Different Directions

- Question
  - Why does one phase boundary move downward while another moves left?
- Evidence
  - Solutions have lower vapor pressure than pure solvent at every temperature.
  - The solid phase remains essentially unchanged by dissolved solute.
- Reasoning
  - $\because$ Only the liquid phase is stabilized, changing every equilibrium involving the liquid but not the pure solid.
  - $\therefore$ Liquid–gas equilibrium requires higher temperature, whereas solid–liquid equilibrium occurs at lower temperature.
- Takeaway
  - One liquid stabilization explains both temperature shifts.

Why Freezing Changes More Than Boiling

- Question
  - Why is freezing-point depression larger than boiling-point elevation for equal molality?
- Evidence
  - For water, $K_f = 1.86$ and $K_b = 0.512\ °\mathrm{C\,kg\,mol^{-1}}$.
  - Water has $\Delta H_{fus} < \Delta H_{vap}$.
- Reasoning
  - $\because$ Smaller phase-change enthalpy means less temperature change restores equilibrium after equal liquid stabilization.
  - $\therefore$ Freezing shifts more than boiling because melting requires much less energy than vaporization.
- Takeaway
  - Phase-change energy controls colligative temperature shifts.

Outline of Part IV: Colligative Properties

Vapor Pressure Lowering

B.P. Elevation and F.P. Depression

Osmosis and Osmotic Pressure

Osmosis and Osmotic Pressure

Osmosis

Diffusion of **solvent molecules** through a **semipermeable membrane** from lower to higher solute concentration.

Osmotic Pressure (Π)

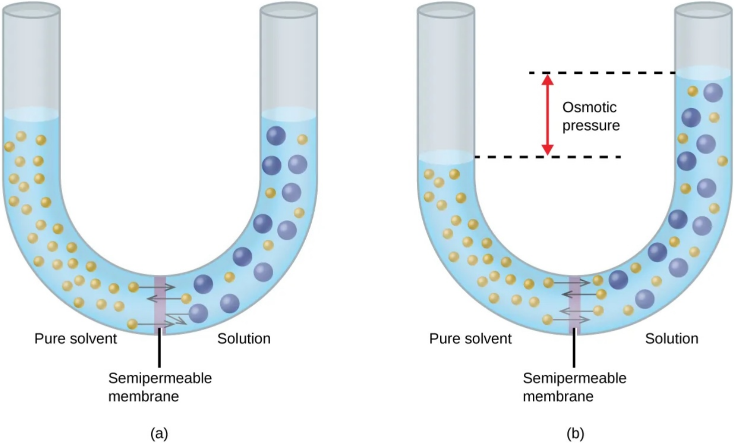
The external pressure required to **stop** net osmosis:

$$\boxed{\Pi = MRT}$$

$M =$ molarity $\left(\frac{\text{mol}}{\text{L}}\right)$

$R = 0.082\,06\ \frac{\text{L atm}}{\text{mol K}}$

$T =$ absolute temperature (K)



(a) Solvent molecules pass through a semipermeable membrane from the pure solvent to the solution, while solute particles cannot cross. (b) The solution level rises until the resulting osmotic pressure balances the net solvent flow, establishing equilibrium $\Pi = MRT$.

What Drives Osmosis?

- Question
  - Why does solvent move toward the more concentrated solution?
- Evidence
  - Solvent crosses a semipermeable membrane until flow eventually stops.
  - Solute remains confined to one side of the membrane.
- Reasoning
  - $\because$ Dissolved solute lowers the solvent *chemical potential*, making additional solvent energetically more stable there.
  - $\therefore$ Net solvent flow continues until both sides reach equal solvent chemical potential.
- Takeaway
  - Chemical-potential differences drive solvent transport.

Why Osmotic Pressure Equals $\Pi = MRT$

- Question
  - How does chemical-potential balance produce the osmotic-pressure equation?
- Evidence
  - Applying pressure raises the solvent chemical potential.
  - Osmotic flow stops at a measurable pressure Π.
- Reasoning
  - $\because$ The applied pressure exactly offsets the lowering caused by dissolved solute, restoring equal solvent chemical potentials.
  - $\therefore$ Dividing the required potential change by solvent molar volume gives $\Pi = cRT$, where c is solute concentration.
- Takeaway
  - Pressure can compensate chemical-potential differences.

Why Osmotic Pressure Resembles the Ideal Gas Law

- Question
  - Why does $\Pi = cRT$ have the same form as the ideal gas law?
- Evidence
  - Dilute solutions obey $\Pi = cRT$.
  - Dilute gases obey $P = cRT$.
- Reasoning
  - $\because$ Dilute particles interact weakly, so each contributes independently to the chemical-potential difference.
  - $\therefore$ Osmotic and gas pressures both become proportional to particle concentration and temperature.
- Takeaway
  - Independent particles create linear pressure laws.

When the Linear Osmotic Law Breaks Down

- Question
  - Why does osmotic pressure become nonlinear at high concentration?
- Evidence
  - Concentrated solutions exceed the prediction of $\Pi = cRT$.
  - Osmotic pressure is highly sensitive for measuring polymer molar masses.
- Reasoning
  - $\because$ Crowding and intermolecular interactions change solvent stabilization beyond simple dilution.
  - $\therefore$ Larger pressure corrections reveal even tiny numbers of dissolved macromolecules.
- Takeaway
  - Interactions control departures from dilute behavior.

How Semipermeable Membranes Select Molecules

- Question
  - How can a membrane pass water yet reject glucose?
- Evidence
  - Water crosses rapidly through many biological membranes.
  - Glucose remains largely excluded under the same conditions.
- Reasoning
  - $\because$ Nanometer-sized pores and selective molecular interactions lower the energy barrier only for appropriately sized molecules.
  - $\therefore$ Solvent moves to equalize chemical potential while larger solutes remain separated.
- Takeaway
  - Selective transport depends on molecular-scale energy barriers.

Osmosis: Applications and Tonicity

Reverse Osmosis

Applying pressure $> \Pi$ forces solvent from the *concentrated* side to the *dilute* side:

- **Seawater desalination**
- Water purification systems

Biological Relevance: Tonicity

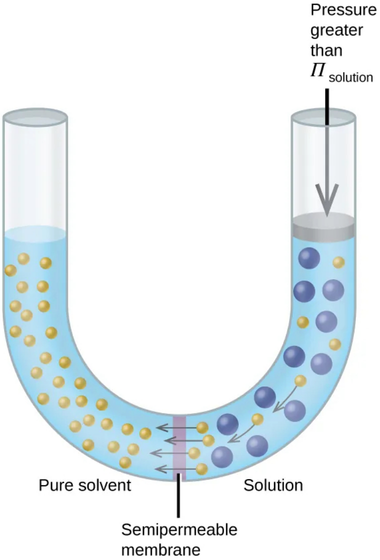
Blood serum $\Pi \approx 7.7\,\text{atm}$.

IV solutions must be **isotonic**:

- **Isotonic**: no net water flow; normal cell volume.
- **Hypotonic**: water flows *into* cells $\Rightarrow$ **hemolysis** (bursting).
- **Hypertonic**: water flows *out* $\Rightarrow$ **crenation** (shriveling).

Osmotic pressure is extremely sensitive to solute concentration and can be measured accurately — it is widely used to determine the molar masses of large biomolecules (proteins, polymers).

Reverse Osmosis: Using Pressure to Purify Solvents



Applying a pressure greater than the solution's osmotic pressure reverses the natural direction of osmosis, forcing solvent through the semipermeable membrane while leaving dissolved solutes behind. Reverse osmosis is widely used for desalination and water purification.

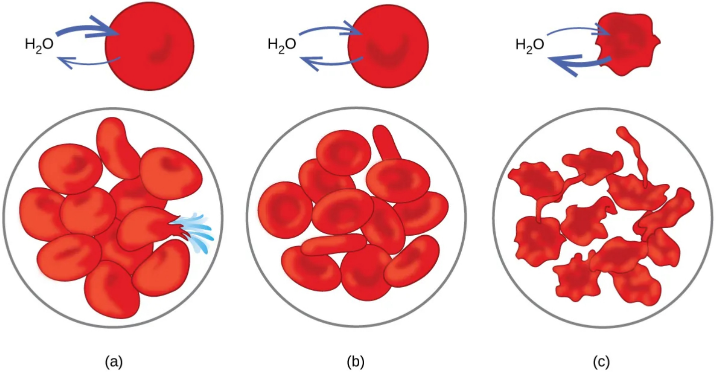
Reverse Osmosis Systems for Water Purification

(a)

(b)

(a) A portable reverse osmosis unit and (b) a large-scale purification system demonstrate how pressure greater than the osmotic pressure forces water through a semipermeable membrane, removing dissolved salts and other impurities to produce clean drinking water.

Effects of Osmosis on Red Blood Cells



(a) In a hypotonic solution, water enters red blood cells, causing them to swell and potentially rupture (hemolysis). (b) In an isotonic solution, water moves equally in both directions, maintaining normal cell size. (c) In a hypertonic solution, water leaves the cells, causing them to shrink (crenation).

Part V

Modification to Colligative Properties

Outline of Part V: Modification to Colligative Properties

Van't Hoff Factor

Outline of Part V: Modification to Colligative Properties

Van’t Hoff Factor

The Van't Hoff Factor for Electrolytes

Van't Hoff Factor (i)

For electrolytes, the actual number of dissolved particles exceeds the number of formula units. The **van't Hoff factor** corrects for this:

$$i = \frac{\text{actual moles of particles in solution}}{\text{moles of formula units dissolved}}$$

Theoretical vs. Actual i

Solute	Ideal i	Actual i (0.05 m)
NaCl	2	≈ 1.9
$MgCl_2$	3	≈ 2.7
Sucrose	1	≈ 1.0

$i <$ theoretical value because of **ion pairing**: residual interionic attraction reduces the effective particle count. At higher dilution, i approaches the ideal value.

Modified equations: $\boxed{\Delta T_b = iK_b m}$ $\boxed{\Delta T_f = iK_f m}$ $\boxed{\Pi = iMRT}$

Why Particle Number Controls Colligative Properties

- Question
  - Why do colligative properties depend on particle number rather than chemical identity?
- Evidence
  - Equal numbers of dissolved particles produce equal ideal osmotic pressures.
  - Nonelectrolytes and fully dissociated salts follow the same proportionality.
- Reasoning
  - $\because$ Each independently moving particle lowers the solvent *chemical potential* by the same dilute-limit amount.
  - $\therefore$ Every independent particle contributes equally, giving colligative properties proportional to effective particle number.
- Takeaway
  - Independent particles determine colligative effects.

Why the Van't Hoff Factor Changes with Concentration

- Question
  - Why does the measured *van't Hoff factor* approach its ideal value upon dilution?
- Evidence
  - Measured i increases as electrolyte concentration decreases.
  - At infinite dilution, measured i approaches the stoichiometric value.
- Reasoning
  - $\because$ Greater ion separation weakens electrostatic attraction, allowing ions to behave as independent particles.
  - $\therefore$ The effective particle count approaches the number predicted by complete dissociation.
- Takeaway
  - Dilution restores independent particle behavior.

How Equilibrium Determines the Van't Hoff Factor

- Question
  - Why does a weak electrolyte have a concentration-dependent van't Hoff factor?
- Evidence
  - Acetic acid dissociates only partially in water.
  - Dissociation increases as the solution becomes more dilute.
- Reasoning
  - $\because$ Dilution shifts the dissociation equilibrium toward separated ions, increasing independently moving particles.
  - $\therefore$ The measured i depends on both the acid's K_a and its concentration.
- Takeaway
  - Equilibrium controls effective particle number.

Why $MgCl_2$ Deviates More Than NaCl

- Question
  - Why is i for $MgCl_2$ farther from its ideal value than for NaCl?
- Evidence
  - Measured i is about 2.7 for $MgCl_2$ and 1.9 for NaCl.
  - Mg^{2+} carries twice the positive charge of Na^+.
- Reasoning
  - $\because$ Larger charge products strengthen electrostatic attraction, increasing transient ion association and correlated motion.
  - $\therefore$ Fewer ions behave independently, reducing the effective particle count further below the ideal value.
- Takeaway
  - Stronger electrostatics reduce particle independence.

Check Your Understanding

Which solution would have the **lowest** freezing point?

(a) 1.0 m glucose ($C_6H_{12}O_6$) — nonelectrolyte

(b) 1.0 m NaCl — strong electrolyte, 2 ions

(c) 1.0 m $MgCl_2$ — strong electrolyte, 3 ions

(d) 0.5 m $MgCl_2$

Come to lecture to see the answer.

Part VI

Colloids

Outline of Part VI: Colloids

Colloidal Dispersions

Outline of Part VI: Colloids

Colloidal Dispersions

What Are Colloids?

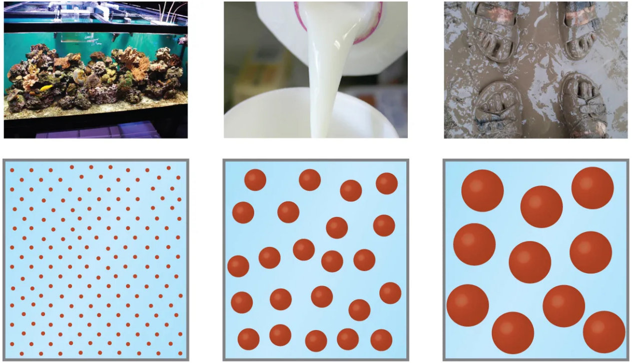
Definition: Colloid (Colloidal Dispersion)

A mixture in which particles are larger than typical solution molecules but small enough that they **do not settle** under gravity.

- **Dispersed phase**: the particulate component.
- **Dispersion medium**: the substance throughout which particles are dispersed.

Type	**Particle Size**	**Settles?**	**Light?**
True solution	$< 1\,\text{nm}$	No	Passes through (transparent)
Colloid	1–1000 nm	No	Scatters (cloudy/opalescent)
Suspension	$> 1000\,\text{nm}$	Yes	Opaque

Solutions, Colloids, and Suspensions: Comparing Particle Size and Behavior



(a) (b) (c)

(a) Solutions contain molecular- or ion-sized particles that remain uniformly dispersed and appear transparent. (b) Colloids contain intermediate-sized particles that stay suspended without settling. (c) Suspensions contain large particles that make the mixture cloudy and eventually settle out under gravity.

Why Colloid Size Defines a New Physical Regime

- Question
  - Why are the solution–colloid and colloid–suspension size boundaries physically meaningful?
- Evidence
  - Molecules smaller than about 1 nm neither settle nor visibly scatter light.
  - Particles larger than about 1000 nm readily sediment under gravity.
- Reasoning
  - $\because$ Increasing particle size changes how strongly gravity and light interact with dispersed matter.
  - $\therefore$ Colloids occupy the crossover where both effects become experimentally observable.
- Takeaway
  - Particle size determines dominant physical behavior.

Why Colloidal Particles Resist Settling

- Question
  - Why does gravity fail to separate most colloidal particles?
- Evidence
  - Colloids remain dispersed for long periods without stirring.
  - Large suspended particles settle even in still liquids.
- Reasoning
  - $\because$ Random thermal collisions continually redirect small particles, overcoming the tiny gravitational energy gained during settling.
  - $\therefore$ Brownian motion dominates until particles become large enough for gravity to prevail.
- Takeaway
  - Competing energies determine suspension stability.

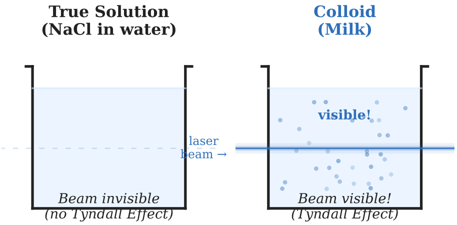
The Tyndall Effect

Colloidal particles are large enough to **scatter visible light**, making a colloid appear **cloudy or opalescent** when a beam passes through it.

Examples

- Light beam visible through fog or smoke.
- A flashlight beam is visible in milk.
- Blue color of the sky (light scattered by atmospheric particles).
- **True solutions** (saltwater, sugar water) show **no** Tyndall Effect.

The Tyndall Effect distinguishes a **colloid** from a **true solution**.



The Tyndall Effect

The paths of searchlight beams are made visible when light is scattered by colloidal-size particles in the air (fog, smoke, etc.).

Why Colloids Scatter Visible Light

- Question
  - Why do colloids show the Tyndall effect while true solutions appear transparent?
- Evidence
  - A light beam is visible through milk but not through salt solution.
  - Visible light has wavelengths of roughly 400–700 nm.
- Reasoning
  - $\because$ Particles approaching the wavelength of light reradiate the oscillating electric field efficiently, whereas much smaller molecules do not.
  - $\therefore$ Colloids strongly scatter visible light, producing the *Tyndall effect*.
- Takeaway
  - Scattering depends on particle–wavelength comparison.

Why the Sky Appears Blue

- Question
  - Why does wavelength-dependent scattering make the sky blue instead of white?
- Evidence
  - Air molecules scatter blue light much more strongly than red light.
  - Sunlight entering the atmosphere is approximately white.
- Reasoning
  - $\because$ Smaller wavelengths interact more strongly with tiny atmospheric particles, increasing the probability of scattering.
  - $\therefore$ Blue light reaches observers from many directions, making the sky appear blue.
- Takeaway
  - Wavelength controls scattering intensity.

Types of Colloidal Systems

Dispersed Phase	Medium	Name	Example
Solid	Gas	Smoke	Smoke, dust
Solid	Liquid	Sol	Paint, starch, milk of magnesia
Solid	Solid	—	Gems, alloys
Liquid	Gas	Aerosol	Clouds, fog, mist
Liquid	Liquid	Emulsion	Milk, mayonnaise
Liquid	Solid	Gel	Jellies, gelatin (Jell-O)
Gas	Liquid	Foam	Whipped cream, sea foam
Gas	Solid	—	Pumice

Preparation of Colloidal Systems

Dispersion Methods

Large particles are **broken into colloidal-sized particles**.

- Mechanical grinding or milling
- Paint pigments
- Aerosol atomizers

Key Idea

Existing particles are reduced to the **1–1000 nm** colloidal size range.

Condensation Methods

Small molecules or ions **aggregate** to form colloidal particles.

- Water vapor $\rightarrow$ cloud droplets
- Chemical reactions produce colloidal solids
- Growth must stop before particles become precipitates

If particles become too large, they settle as a **precipitate** rather than forming a colloid.

Emulsions and Examples of Colloid Formation

Emulsions

An **emulsion** is a colloidal dispersion of one liquid in another immiscible liquid.

- Formed by shaking or blending.
- Droplets tend to merge (coalesce).
- Stabilized by an **emulsifying agent**.

Common Examples

- **Milk**: butterfat in water (casein = emulsifier)
- **Mayonnaise**: oil in vinegar (egg yolk = emulsifier)
- **Soap**: stabilizes oil–water emulsions
- **Clouds**: condensed water droplets in air

Emulsions and Emulsifying Agents

Emulsifying Agent

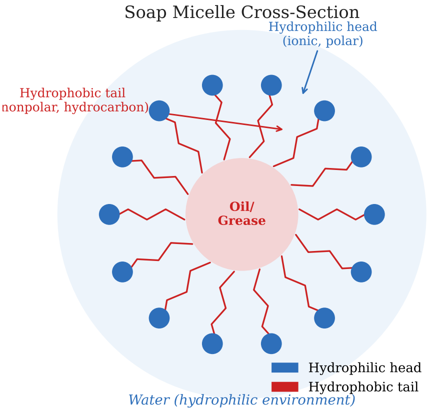
An **amphiphilic** molecule with both a **hydrophobic** (nonpolar) end and a **hydrophilic** (polar/ionic) end.

How It Works

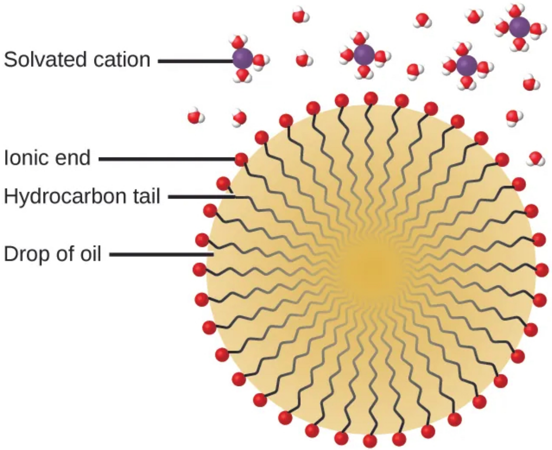
- Nonpolar tails embed in oil droplets.
- Polar/ionic heads face the water.
- This prevents droplets from coalescing.

Examples

- **Milk**: butterfat in water; stabilized by **casein protein**.
- **Mayonnaise**: oil in vinegar; stabilized by **lecithin** from egg yolk.



Emulsions and Emulsifying Agents



Soap and detergent molecules **stabilize** oil droplets in water.

The nonpolar hydrocarbon tails embed in the oil droplet, while the polar ionic heads remain in contact with the surrounding water. This amphiphilic arrangement forms a protective coating around the oil, allowing it to remain dispersed (emulsified) in water so it can be washed away.

Solvated cations are shown associated with the ionic head groups.

Soaps and Detergents

Soap

Long nonpolar hydrocarbon chain + ionic **carboxylate** group ($-COO^-$).

Example: Sodium stearate, $C_{17}H_{35}CO_2Na$

Detergent

Similar amphiphilic structure but with a **sulfate** ($-OSO_3^-$) or **sulfonate** ($-SO_3^-$) head group instead of carboxylate.

Why Detergents Beat Soaps in Hard Water

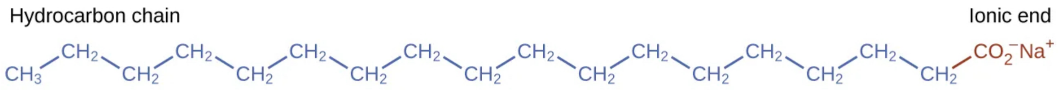
Hard water contains Ca^{2+} and Mg^{2+} ions. Soaps form **insoluble precipitates** ("soap scum") with these ions, reducing cleaning power.

Detergent sulfonate/sulfate groups form **water-soluble** products even with Ca^{2+} and Mg^{2+}.

Cleaning Mechanism

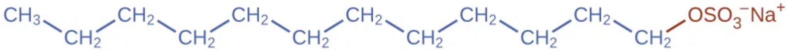
Nonpolar tails surround grease/dirt; ionic heads face water; grease is suspended as a colloidal dispersion and rinsed away.

Soaps and Detergents



sodium stearate (soap)

Soaps contain a nonpolar hydrocarbon end (blue) and an ionic end (red). The ionic end is a carboxylate group. The length of the hydrocarbon end can vary from soap to soap.



sodium lauryl sulfate (detergent)

Detergents contain a nonpolar hydrocarbon end (blue) and an ionic end (red). The ionic end can be either a sulfate or a sulfonate. The length of the hydrocarbon end can vary from detergent to detergent.

Why Amphiphiles Spontaneously Form Micelles

- Question
  - Why does the hydrophobic effect create ordered micelles instead of dispersed molecules?
- Evidence
  - Amphiphiles self-assemble above a characteristic concentration.
  - Oil droplets become surrounded by amphiphiles in water.
- Reasoning
  - $\because$ Isolated hydrocarbon tails force nearby water into constrained hydrogen-bond arrangements, while clustering releases many water molecules.
  - $\therefore$ Increased water entropy outweighs amphiphile ordering, making *micelles* thermodynamically favorable.
- Takeaway
  - Water entropy drives hydrophobic self-assembly.

Why Micelles Form Above a Critical Concentration

- Question
  - Why does micelle formation begin abruptly at the critical micelle concentration?
- Evidence
  - Few micelles exist below the critical micelle concentration.
  - Above it, additional amphiphiles mainly enter existing or new micelles.
- Reasoning
  - $\because$ Small aggregates expose many hydrocarbon tails, giving too little water-entropy gain to offset aggregation costs.
  - $\therefore$ Once aggregates become large enough, adding molecules lowers free energy cooperatively.
- Takeaway
  - Stable self-assembly begins beyond an energetic threshold.

Why Detergents Work in Hard Water

- Question
  - Why do detergents clean effectively in hard water while soaps do not?
- Evidence
  - Soap forms insoluble deposits with Ca^{2+} and Mg^{2+}.
  - Most detergents remain dissolved under the same conditions.
- Reasoning
  - $\because$ Carboxylate head groups form much less soluble calcium salts than sulfonate or sulfate head groups.
  - $\therefore$ Detergents retain dispersed amphiphiles that continue forming micelles in hard water.
- Takeaway
  - Head-group solubility controls cleaning performance.

Why Micelles Have a Preferred Size

- Question
  - Why do micelles stop growing instead of forming one giant aggregate?
- Evidence
  - Micelles have reproducible sizes under fixed solution conditions.
  - Larger aggregates contain more closely packed charged head groups.
- Reasoning
  - $\because$ Hydrophobic attraction favors growth, whereas electrostatic repulsion between head groups increases with crowding.
  - $\therefore$ Micelles stabilize where these opposing energetic effects balance.
- Takeaway
  - Self-assembly reflects competing molecular interactions.

Electrical Properties

Charge Stability of Colloids

All colloidal particles in a given system carry the **same sign of charge**:

- Metal hydroxide colloids: typically *positive*.
- Metal and metal sulfide colloids: typically *negative*.

Electrostatic repulsion between like-charged particles **prevents aggregation and settling**.

Adding an electrolyte to a colloid can cause **coagulation**: the added ions neutralize the surface charges, removing the repulsive barrier and allowing particles to aggregate and settle.

Case Study: Deepwater Horizon (2010)

Dispersant Corexit emulsified 4.9 million barrels of oil into colloidal droplets, increasing surface area for bacterial degradation.

Check Your Understanding

A student shines a laser pointer into two cups:

(A) Saltwater (NaCl dissolved in water)

(B) A glass of milk

In which cup is the laser beam **visible from the side**, and why?

Come to lecture to see the answer.

Cottrell Precipitator: Removing Colloidal Particles from Air

Electrostatic Precipitator

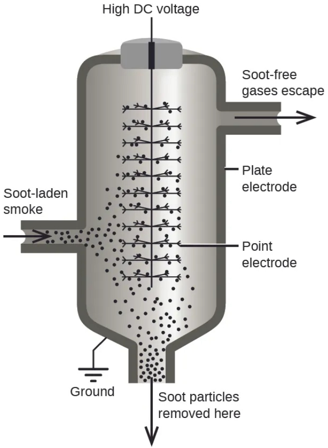
A Cottrell precipitator removes fine colloidal particles from gases using a strong electric field.

Operation

- Particles in polluted air become electrically charged.
- Charged particles are attracted to oppositely charged electrodes.
- Particles are neutralized and collected as dust.

Applications

- Power plant emissions
- Cement and steel production
- Industrial air pollution control



Electrostatic precipitation removes smoke, dust, and other colloidal particles from exhaust gases by electrically collecting suspended particles.

Why Do Colloidal Particles Acquire Surface Charge?

- Question
  - Why do dispersed colloidal particles develop the same surface-charge sign?
- Evidence
  - Colloids commonly contain charged surface groups or selectively adsorb solution ions.
  - Stable colloids show particles with one dominant charge sign.
- Reasoning
  - $\because$ Surface chemistry favors one charge-forming process, every particle gains similar *surface charge*.
  - $\therefore$ Like-charged particles repel before touching, creating a common interaction mechanism.
- Takeaway
  - Surface chemistry creates uniform electrostatic interactions.

Why Do Some Colloids Remain Dispersed?

- Question
  - Why do charged particles stay separated instead of permanently sticking together?
- Evidence
  - All particles attract through *van der Waals* interactions.
  - Stable colloids resist aggregation despite continual collisions.
- Reasoning
  - $\because$ Electrostatic repulsion raises particle potential energy before attractive forces dominate.
  - $\therefore$ Thermal collisions rarely overcome the energy barrier, so dispersion persists.
- Takeaway
  - Stability depends on competing attractive and repulsive interactions.

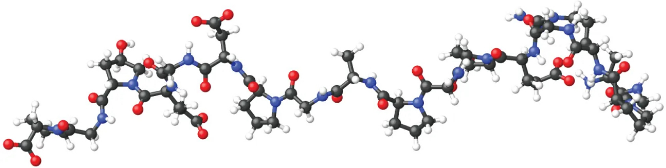
Why Does Adding Electrolyte Cause Coagulation?

- Question
  - How does dissolved NaCl remove the repulsive barrier between particles?
- Evidence
  - Increasing electrolyte concentration rapidly promotes coagulation.
  - Counterions accumulate near oppositely charged particle surfaces.
- Reasoning
  - $\because$ Nearby counterions partially shield electric fields, shortening the *Debye length*.
  - $\therefore$ Repulsion acts over shorter distances, allowing van der Waals attraction to bind particles.
- Takeaway
  - Ion screening lowers the electrostatic energy barrier.

How Does a Cottrell Precipitator Destabilize a Colloid?

- Question
  - Why does a strong electric field remove suspended particles from gases?
- Evidence
  - Charged particles migrate toward oppositely charged collection plates.
  - Industrial precipitators efficiently remove smoke and dust.
- Reasoning
  - $\because$ The electric field separates particles from one another by driving them onto conducting plates.
  - $\therefore$ Removing stabilizing charge from suspension eliminates electrostatic stabilization and particles collect.
- Takeaway
  - External fields can eliminate electrostatic colloid stabilization.

Gels: Colloidal Solid Networks



Gelatin desserts are examples of gels, where a liquid phase containing water, sweeteners, and flavors is trapped within a three-dimensional network of solid gelatin proteins. As gelatin cools, the protein structure forms a stable colloidal gel that gives the dessert its semisolid texture.

What Makes a Gel Different from a Liquid?

- Question
  - Why can two materials containing mostly liquid behave so differently?
- Evidence
  - A liquid flows because its particles move past one another.
  - A gel contains mostly liquid but keeps its overall shape.
- Reasoning
  - $\because$ A gel contains many connected particles spread throughout the liquid.
  - $\therefore$ The liquid remains present, but the connected particles limit large-scale flow.
- Takeaway
  - Gels are liquids supported by connected particles.

Why Do Some Colloids Form Gels?

- Question
  - Why do tiny particles sometimes join into one large structure instead of staying separate?
- Evidence
  - Tiny particles are always moving and colliding.
  - Some mixtures become jelly-like without becoming completely solid.
- Reasoning
  - $\because$ Some collisions create attractions strong enough to keep particles together.
  - $\therefore$ More collisions gradually build one connected structure throughout the liquid.
- Takeaway
  - Strong attractions build connected particle structures.

When Do Particles Stay Together?

- Question
  - Why do some particle collisions last while others do not?
- Evidence
  - Some particles separate immediately after colliding.
  - Others remain attached after they collide.
- Reasoning
  - $\because$ Particles separate only if random motion can overcome their attraction.
  - $\therefore$ Stronger attractions produce longer-lasting particle connections.
- Takeaway
  - Attraction strength controls connection lifetime.

Why Can a Gel Act Like a Solid?

- Question
  - How can something that is mostly liquid resist being pushed or stretched?
- Evidence
  - Gels contain much more liquid than solid material.
  - A gentle push changes their shape only temporarily.
- Reasoning
  - $\because$ The connected particles pull on one another throughout the liquid.
  - $\therefore$ The whole material resists changing shape even though most of it is liquid.
- Takeaway
  - Connected particles make liquids resist deformation.

Why Do Only a Few New Connections Matter?

- Question
  - Why can adding only a few particle connections make a gel much stronger?
- Evidence
  - A gel becomes much firmer over a short time.
  - Most possible particle contacts never form.
- Reasoning
  - $\because$ New connections join smaller groups into one larger connected structure.
  - $\therefore$ The entire material becomes harder to deform.
- Takeaway
  - Large connected structures determine material strength.

Why Does Gelatin Form a Gel When It Cools?

- Question
  - What changes in gelatin molecules when the mixture cools?
- Evidence
  - Warm gelatin flows like a liquid.
  - Cool gelatin becomes soft and holds its shape.
- Reasoning
  - $\because$ Cooling allows parts of nearby protein molecules to stick together more easily.
  - $\therefore$ Many proteins become connected into one three-dimensional structure.
- Takeaway
  - Molecular attractions create large connected structures.

Why Does Cooling Help a Gel Form?

- Question
  - Why does cooling help particles stay connected instead of only slowing them down?
- Evidence
  - Molecules continue moving after cooling.
  - Particle connections last longer at lower temperatures.
- Reasoning
  - $\because$ Slower particles have less energy to pull away from attractive neighbors.
  - $\therefore$ Existing connections break less often than new ones form.
- Takeaway
  - Lower temperature favors lasting attractions.

Why Does a Gel Form Instead of a Solid Lump?

- Question
  - Why do some particle mixtures become gels instead of forming a solid lump?
- Evidence
  - Some mixtures stay evenly spread through the liquid.
  - Others collect into one dense mass.
- Reasoning
  - $\because$ Widely distributed particle connections can form before particles collect into one place.
  - $\therefore$ The liquid becomes a gel instead of producing a separate solid.
- Takeaway
  - Evenly distributed connections produce gels.

Summary: Chapter 11 — Solutions and Colloids

- A **solution** is a homogeneous mixture; the solvent sets the phase. Dissolution is driven by energy minimization and/or entropy increase.
- **Electrolytes** produce ions (strong: complete; weak: partial); **nonelectrolytes** produce none.
- **Solubility**: "like dissolves like"; gas solubility decreases with T and follows **Henry's Law** ($C_g = kP_g$) for pressure; most solid solubilities increase with T.
- **Colligative properties** depend only on total particle concentration:
  - Vapor pressure lowering (Raoult's Law: $P_A = X_A P_A^*$)
  - BP elevation ($\Delta T_b = K_b m$); FP depression ($\Delta T_f = K_f m$)
  - Osmotic pressure ($\Pi = MRT$)
  - For electrolytes, multiply by van't Hoff factor i.
- **Colloids** (1–1000 nm particles) scatter light (Tyndall Effect), do not settle, and are used in soaps, emulsions, and industrial precipitators.